

FINAL PROJECT SUBMISSION

—
MScFE Capstone Project

Signal Portfolio Construction via Modified Determinant Model

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Abstract

We study how correlations among seven major global equity indices, SP 500, B3-Bovespa, EURO STOXX 50, FTSE 100, FTSE/JSE Top 40, Nikkei 225, and ASX 50, change across time horizons and market conditions, and whether that structure can support better portfolio construction. Our sample study runs from 2015 to 2025. A recurring problem in global correlation work is that non-overlapping trading hours make daily co-movements look smaller than they really are. We compute correlations at horizons from one day to four weeks and find that the mean pairwise correlation rises from 0.252 at the daily level to 0.547 at four weeks.

To filter noise from the dependence estimates, we introduce KIKO - (Koherent Interdependence Korrelation Operator) to study the Interdependence Structure via Adaptive Asset Correlations (ISAAC). We track the modified determinant $R - I$ of the rolling correlation matrix across 3-, 7-, and 28-day windows to extract Buy-or-Sell signals for the underlying assets. On the portfolio side, our Interdependence World Asset Networks framework (called I-WAN) first picks the best-performing strategy for each individual index and then assembles those into four global overlays. One result that deserves attention is that during the COVID-19 sell-off in March 2020, all four global portfolios fell 33% peak-to-trough, suggesting that cross-country diversification did not provide the protection it appeared to offer in calmer periods. These findings laid the ground for researching alternative measures and metrics to overcome that problem. Throughout our study, we show how the determinant signal can lead to improved Sharpe ratios and overall performance compared to various benchmark portfolio strategies.

Keywords: market correlations, multi-scale analysis, portfolio optimization, regime detection, global equity markets, hierarchical risk parity, non-synchronous trading, diversification

⁰https://github.com/kboroz/MScFE_Capstone_Project/tree/main

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List of Abbreviations

Abbreviation	Description
ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller
ASX	Australian Securities Exchange
BH	Buy-and-Hold
CVaR	Conditional Value at Risk
EDA	Exploratory Data Analysis
EW	Equal-Weight
EW-BH	Equal-Weight Buy-and-Hold
EW-M	Equal-Weight Monthly Rebalancing
EW-Q	Equal-Weight Quarterly Rebalancing
FX	Foreign Exchange
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
HRP	Hierarchical Risk Parity
IQR	Interquartile Range
ISAAC	Interdependence Structure via Adaptive Asset Correlations
I-WAN	Interdependence World Asset Network
KIKO	Koherent Intermarket Korrelation Optimizer
MCD	Minimum Covariance Determinant
ML	Machine Learning
MDD	Maximum Drawdown
MST	Minimum Spanning Tree
MV	Mean-Variance
MV-BH	Mean-Variance Buy-and-Hold
MV-M	Mean-Variance Monthly Rebalancing
MV-Q	Mean-Variance Quarterly Rebalancing
PACF	Partial Autocorrelation Function
RMT	Random Matrix Theory
VaR	Value at Risk
VIX	CBOE Volatility Index

1 Introduction

A problem in global correlation networks is that non-overlapping trading hours make daily co-movements look smaller than they really are. We study how correlations among seven major global equity indices, S&P 500, B3-Bovespa, EURO STOXX 50, FTSE 100, FTSE/JSE Top 40, Nikkei 225, and ASX 50, change across time horizons and market conditions, and whether that structure can support better portfolio construction. Portfolio theory has always rested on correlation estimates. Since Markowitz (1952), the covariance matrix has been the input that determines how much diversification a portfolio actually achieves. The problem is that the covariance matrices that analysts compute from real return data are noisy. Laloux et al. (1999) and Plerou et al. (1999) showed, independently and around the same time, that most of the eigenvalue structure in an empirical financial correlation matrix is a statistical artifact. An optimizer fed that matrix will construct a portfolio that looks efficient on historical data and often falls apart out of sample. That noise problem is well-known. Less discussed is a separate distortion that affects global portfolios specifically: non-synchronous trading hours. The S&P 500 and Nikkei 225 barely overlap during a trading day. When computing daily return correlations, you are comparing prices set hours apart, in different information environments. The measured correlation is lower than the true economic co-movement between the two markets, not because the markets are genuinely independent, but because the daily differencing period is too short to capture the lag in information transmission. Extend the horizon to a week, and the artifact largely disappears. Despite how obviously this matters for international portfolio construction, there is surprisingly little systematic evidence on how correlations change across horizons for a geographically diverse set of indices that spans multiple time zones, and both developed and emerging markets. The second problem is regime instability. Correlations between equity markets are not stable. During periods of stress, they converge. Forbes and Rigobon (2002) warned that the apparent contagion seen during crises partly reflects a mechanical inflation of correlation estimates under heteroskedasticity, but even after correcting for that, co-movements demonstrably increase when volatility spikes. Preis et al. (2012) quantified this directly for S&P 500 stocks. The practical consequence is uncomfortable: the diversification benefits that justify global allocation in normal markets shrink or disappear in the conditions where you most need them. March 2020 made this vivid. Every equity market in our universe fell together, with correlation approaching one, regardless of what a pre-COVID covariance matrix would have predicted. This paper addresses both problems. We cover seven indices, S&P 500, B3-Bovespa, EURO STOXX 50, FTSE 100, FTSE/JSE Top 40, Nikkei 225, and ASX 50, from January 2015 through December 2025. Three contributions organize the work. First, a multi-scale correlation analysis across horizons from one day to four weeks, using Pearson, Spearman, Kendall, and copula measures. This directly quantifies how the non-synchronous trading artifact decays with the return horizon and what the horizon-specific clustering structure looks like across the seven markets. The final report is structured as follows: Section 2 reviews the literature. Section 3 describes the necessary data and preprocessing steps. Section 4 covers the exploratory analysis. In Section 5 we present benchmark portfolio constructions and their performances. Section 6 focusses on the ISAAC regime detection framework and covers correlation denoising via the KIKO framework. Section 7 closes our report and consolidates the results.

2 Literature Review

2.1 Noise in Empirical Correlation Matrices

The standard sample correlation matrix, estimated from a finite time series of asset returns, is unreliable. When the number of assets is large relative to the number of observations, as is a typical situation in portfolio construction, many off-diagonal entries reflect sampling noise rather than genuine economic co-movement. Feed this noisy matrix into a Markowitz optimizer, and it will find weights that exploit the most extreme spurious correlations, building a portfolio that looks efficient in-sample and frequently fails out-of-sample. This is a known and persistent failure mode, not an edge case. Laloux et al. (1999) and Plerou et al. (1999) applied Random Matrix Theory (RMT) to this problem. The key insight is that the eigenvalue distribution of a purely random matrix has a known theoretical form (the Marchenko-Pastur law), determined by the ratio of assets to observations. When you compare the empirical eigenvalue spectrum of a financial correlation matrix against this theoretical baseline, the vast majority of eigenvalues fall inside the random band. Only a handful exceed it. Those are the ones carrying actual economic signals, the broad market factor, a few sector factors, perhaps some regional effects. The rest is noise. The practical fix is to replace the noise eigenvalues with their theoretical mean before inverting the matrix, which produces a denoised estimate that gives more stable portfolio weights. Tumminello, Lillo, and Mantegna (2010) took a different angle, using graph-theoretic tools such as Minimum Spanning Trees and planar maximally filtered graphs to extract the dependence hierarchy latent in financial correlation matrices. These network representations make the structure of market co-movements visible in a way that a heatmap does not: you can see which assets are the most connected hubs, which are peripheral, and which clusters share a common driver. Kubiak et al. (2024) pushed further into machine learning territory, proposing Denoising Diffusion Probabilistic Models to generate synthetic correlation matrices that preserve empirical structure while filtering noise. The RMT approach remains the standard in this literature, but the diffusion model work is a sign of where the field is going. The Minimum Covariance Determinant (MCD) estimator offers yet another route. Rather than correcting a noise-contaminated sample covariance matrix after the fact, the MCD builds a robust estimate by finding the subset of observations with the smallest covariance determinant. It is less sensitive to outliers and fat-tailed return episodes, which is exactly the kind of robustness needed for financial data.

2.2 Multi-Scale Analysis and Non-Synchronous Trading

Markets in different time zones are usually not open at the same time, so their daily closing prices reflect information from various sources of the day. The S&P 500 close comes hours after the Nikkei close, and the JSE Top 40 close falls between the two. When you compute a daily return for each and correlate them, you are not measuring how the same information flows move both markets simultaneously. You are measuring two separate days. The resulting correlation is artificially depressed. Fernandez-Macho (2012) documented the scale-dependence of this effect for European stock markets using a wavelet multiple correlation framework. The wavelet decomposition separates a return series into frequency-specific com-

ponents, making it possible to estimate correlations at daily, weekly, and monthly timescales from the same data. European markets looked much more integrated at weekly and monthly frequencies than at the daily level. The short-horizon attenuation was consistent with microstructure effects and non-synchronous trading rather than genuinely lower economic linkage. Michis (2022) added partial correlations to this framework. Conditioning on the overall market factor at each frequency reveals that many apparent pairwise co-movements at short horizons are not direct relationships at all, just shared exposure to the same common driver. The direct bilateral links become clearer at longer horizons once the common factor is partialled out. This has a practical implication: short-horizon correlations can overstate or understate the actual economic connection between two markets, depending on whether you are picking up direct linkages or factor contamination. Ren et al. (2021) examined how these structures shift across market regimes in a global setting, finding that the correlation network among equity markets becomes substantially denser during stress periods. Song et al. (2011) showed similar effects using correlation-based graph analysis over a longer historical sample, with notable network tightening during the 2008 financial crisis. Both papers reinforce the same basic conclusion: the correlation structure you observe in calm markets is not the one you get during drawdowns.

2.3 Copula Models and Tail Dependence

Pearson correlation has a specific weakness that matters a great deal for risk management. It measures average linear co-movement. It says nothing about whether markets tend to crash together harder than they rally together, or whether the tails of the joint return distribution are fatter than a bivariate normal would predict. For a portfolio manager primarily concerned with extreme losses, these are not minor technical details. They are the whole point. Copula models address this directly. By Sklar’s theorem, a multivariate distribution can be separated into its marginal distributions and a copula that captures the full dependence structure. This means you can model fat-tailed or skewed marginals for each asset independently, then specify a copula that allows for asymmetric or heavy-tailed dependence, without forcing everything into the constraints of multivariate normality. Czado (2019) and Hofert et al. (2018) cover the framework comprehensively; vine copulas, which build high-dimensional dependence structures through sequences of bivariate copulas arranged in a tree, are particularly useful for larger asset universes because they are flexible without being intractable. Applied to forecasting, Mwai, Omari, and Mundia (2025) showed that a multivariate GARCH-vine copula combination outperforms standard GARCH with normal or Student-t innovations in tail risk estimation. In our work, copula-based correlation measures serve as a robustness check alongside Pearson and Spearman. The finding that mean pairwise copula correlation rises from 0.252 at the daily horizon to 0.547 at four weeks mirrors the linear correlation results, confirming that the multi-scale effect is not an artifact of the linear correlation assumption.

2.4 Dynamic Correlation and Volatility Clustering

Return volatility clusters. Large moves tend to be followed by further large moves before eventually subsiding, a feature that GARCH models were designed to capture at the uni-

variate level. Engle’s (2002) Dynamic Conditional Correlation (DCC) model extended this to the multivariate case, allowing correlations between assets to vary over time alongside conditional volatility. The DCC correlation estimates tend to rise during high-volatility periods and fall back during quiet ones, which matches the empirical observation that market co-movements intensify during stress. Preis et al. (2012) measured this directly for S&P 500 stocks, showing that the distribution of pairwise correlations narrows and shifts upward during market downturns: stocks that normally correlate modestly start moving nearly in lockstep. Bae and Elkamhi (2021) found a similar dynamic operating across international markets through a single global correlation factor that spikes during systemic stress episodes. These results are not just descriptive. They imply that a portfolio built using a historical covariance matrix calibrated in normal conditions will be less diversified than it appears when a shock hits. Our ISAAC framework takes a different approach to capturing this dynamic. Rather than fitting a DCC model, we track the determinant of the correlation matrix over rolling windows. The determinant reduces the multidimensional co-movement structure into a single number via identifying when the market is transitioning between decoupled and coupled regimes by changing its sign and trajectory across multiple time windows.

2.5 Network Approaches and Hierarchical Clustering

The Minimum Spanning Tree, introduced to financial markets by Mantegna (1999), was an early attempt to extract meaningful structure from a noisy pairwise correlation matrix without inverting it. The MST connects all assets in a tree that preserves the strongest correlations while removing redundant links, producing a map of which assets are most central to the co-movement network, and which sit at the periphery. It is more interpretable than the full correlation matrix and reveals clustering patterns that are not obvious from a heatmap. Hierarchical clustering generalizes this. Applied to a return correlation or distance matrix, it groups assets by similarity of return behavior through an agglomerative process that produces a dendrogram, a tree showing which assets merge into clusters first and at what correlation level. Assets sharing a tight cluster are essentially substitutes from a diversification standpoint; assets in different branches of the tree offer more genuine risk reduction. Lopez de Prado’s Hierarchical Risk Parity (HRP) method turns this dendrogram directly into portfolio weights. Rather than inverting the covariance matrix, which amplifies estimation error in high-dimensional settings, HRP uses recursive bisection along the hierarchical structure to allocate risk. Assets in tight clusters receive lower combined weight because they are substitutable; assets in distinct branches get proportionally more. Michis (2022) showed that HRP produces more stable out-of-sample performance than mean-variance optimization for large asset universes, which is the setting where covariance inversion is most problematic. Narayan et al. (2023) provided evidence that correlation-based portfolio tilts add value over long horizons for international investors. Strategies that overweight low-correlation assets delivered better risk-adjusted returns than static benchmarks, suggesting the correlation structure has some predictive content that markets do not fully price.

2.6 Long-Run Global Correlations and Crisis Contagion

Goetzmann, Li, and Rouwenhorst (2001) studied international equity correlations back to the nineteenth century and found that they do not trend monotonically with globalization. Correlations were comparably high during the pre-World War I era of capital market integration. They rise and fall with political and institutional conditions, not just trade and investment flows. This historical perspective matters because it pushes back against the assumption that today’s high global co-movement is a structural feature that will only increase. It may be partly cyclical. Forbes and Rigobon (2002) identified a methodological problem in the crisis contagion literature. Pearson correlations are often inflated during high-volatility periods because of heteroskedasticity. When correcting for this, much of what looks like contagion during crises turns out to be a continuation of pre-existing co-movement, not a structural break. The implications are manifold: if crisis co-movement reflects stable interdependence rather than anomalous contagion, diversification across correlated markets cannot save investment positions during a sell-off and must be supplemented by active hedging. Shahzad et al. (2014) documented a specific asymmetry in emerging market correlations. Markets like the Ibovespa and JSE Top 40 show lower average co-movement with developed markets in normal conditions, which makes them look attractive from a diversification standpoint. During global stress events, however, their correlation with developed markets rises disproportionately. The diversification benefit is largest precisely when markets are calm, and you have no particular reason to want it, and smallest during sell-offs when you do.

3 Data

3.1 Data Sources and Acquisition and Preprocessing

The price data for the constituent stocks within each equity index were sourced from Yahoo Finance and stored in our GitHub repository. The dataset consists of index-level constituent price data and a macro dataset.

- **Index dataset:** Daily adjusted closing prices for all stocks within each index.
 - S&P 500: 460 stocks
 - Ibovespa: 56 stocks
 - EURO STOXX 50: 46 stocks
 - FTSE 100: 90 stocks
 - JSE Top 40: 31 stocks
 - Nikkei 225: 217 stocks
 - ASX 50: 44 stocks

-
- **Macro dataset:** Daily adjusted closing prices for all assets:
 - Equity indices (S&P 500, Ibovespa, EURO STOXX 50, FTSE 100, JSE Top 40, Nikkei 225, ASX 50)
 - FX rates (EUR/USD, GBP/USD, JPY/USD, AUD/USD, BRL/USD, ZAR/USD)
 - 10-year government bond yields (US, UK, Germany, Australia, South Africa, Japan)
 - Commodity prices (Gold and WTI Crude Oil)
 - Bitcoin
 - VIX

All data are aligned to a common trading calendar spanning January 2015 to December 2025. Minor gaps resulting from public holidays are forward-filled, while observations with excessive missing data are excluded. We calculated log returns from adjusted closing prices as

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right), \quad (1)$$

where P_t denotes the adjusted closing price at time t . All time series are aligned to a common date index. Index-level constituent data and macro-level asset data are stored separately to support two distinct correlation structures: within-index correlations (intra-correlations) used for regime identification and cross-asset correlations used for portfolio allocation (inter-correlations).

Table 1: Dataset Inventory - Table 2.1: Performance metrics used throughout the study.

Dataset	Rows	Columns	Date Range	Source
SP500	2,413	460	2015-01-05 to 2025-12-30	Yahoo Finance
Ibovespa	2,413	56	2015-01-05 to 2025-12-30	Yahoo Finance
EUROSTOXX50	2,413	46	2015-01-05 to 2025-12-30	Yahoo Finance
FTSE100	2,413	90	2015-01-05 to 2025-12-30	Yahoo Finance
JSETop40	2,413	31	2015-01-05 to 2025-12-30	Yahoo Finance
NIKKEI225	2,413	217	2015-01-05 to 2025-12-30	Yahoo Finance
ASX50	2,413	44	2015-01-05 to 2025-12-30	Yahoo Finance
Macro	2,413	23	2015-01-05 to 2025-12-30	Yahoo Finance / FRED

Note: All eight datasets share identical index lengths after alignment.

Bond yields are differenced and scaled to approximate bond price returns. A fixed modified duration of 7 years is used for all six sovereign bonds, which is a rough but reasonable assumption for 10-year instruments. Yield changes above the 99th percentile are winsorized to prevent single-day data errors from distorting the results.

4 EDA - Exploratory Data Analysis

Before starting our portfolio optimization and the determinant signal part, we ran an extensive exploratory data analysis to study the underlying time series features. We tested all time series for stationarity with different specifications and lag orders through the Augmented Dickey–Fuller test. Here, we kindly ask the reader to inspect our code in our GitHub repository for more specific results. We decided not to include all possible econometric results here in order to focus on our original idea.

Below, we highlight only the distributional fits of the log-return distributions for all indices and typical financial time-series characteristic plots in Figure 2, which is shown here as an example for the S&P 500 index. We observe the standard behavior of the autocorrelation and partial autocorrelation functions across all given indices. This preliminary feature analysis would lay the path for incorporating generalized autoregressive conditional heteroscedasticity (GARCH) models. Due to the extensive simulation and portfolio optimization runs, we decided not to further follow this route and leave it as a potential extension of our approach.

In Figure 3, we plot the pairwise rolling correlations for daily and weekly periods. One can easily observe the very oscillatory and non-stationary behavior of these rolling correlations. This is apparent for each individual combination within every large index as well as across indices. In Figure 4, we observe how the rolling correlation can be viewed as an aggregated measure over time. On short timescales, for example 2 or 3 days, we clearly observe the highly oscillatory behavior of the correlation, as markets are either moving up or down on consecutive days, leading to a positive or negative sign in the correlation. If one averages out this effect over longer time periods, the correlation distribution becomes more representative, as depicted for the two-, three-, and four-week horizons. We would like to highlight and emphasize that correlation should not be viewed as a static metric but rather as a dynamic one. In Figure 5, we display the average rolling correlation from our copula correlation analysis code. Given our global portfolio, we decided not to choose the shortest possible timescale as a leading feature and instead extended the analysis toward mid-range and long-range horizons. By calculating the differences between the correlation plots, we looked for a regime in which the changes become small. By taking a three-day rolling correlation as our shortest signal indicator, we assume that the information transmission from Asian to American markets, with roughly a 16- to 18-hour time span, may cause potential short-term signals to disappear. We argue that if dependencies are changing across the global market structure, a three-day period provides a more robust measure, and therefore we decided to use the three-day period. When checking the differences in Figure 6 between the five- and seven-day periods, almost no large changes are present. We therefore decided to take that range for our mid-term signal. A 28-day horizon was chosen to incorporate a long-term signal into our estimates. All other combinations are possible, but due to the extensive calculation times, especially in the determinant calculation part, and the factorial growth of all possible combinations, we decided to use the 3-day, 7-day, and 28-day windows throughout our analysis.

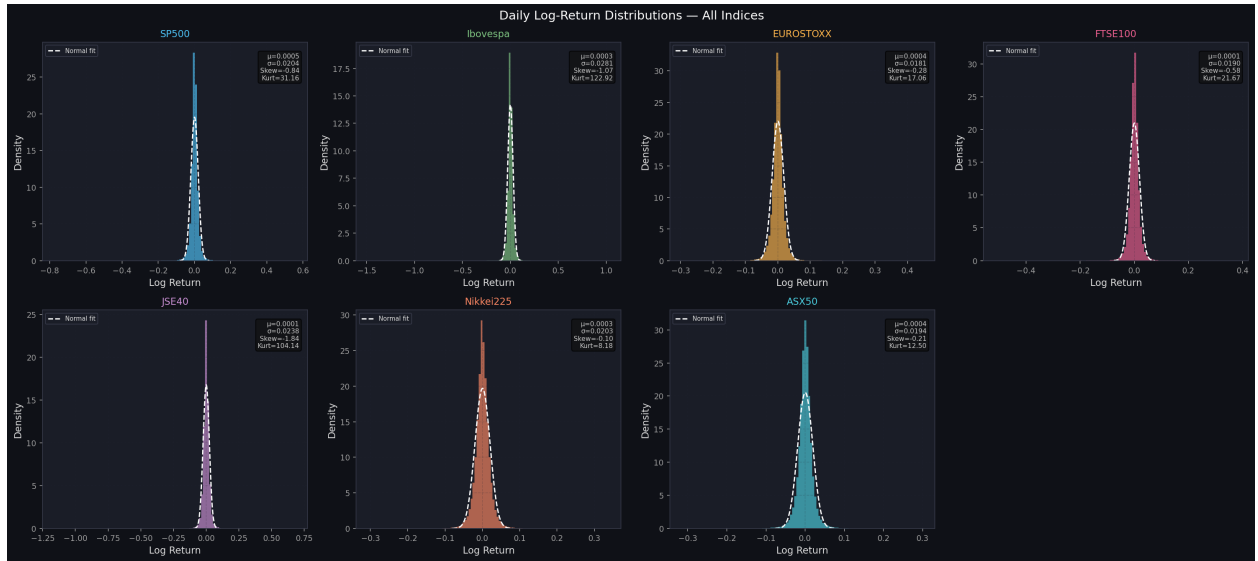


Figure 1: Daily Log-Return Distributions — All Indices

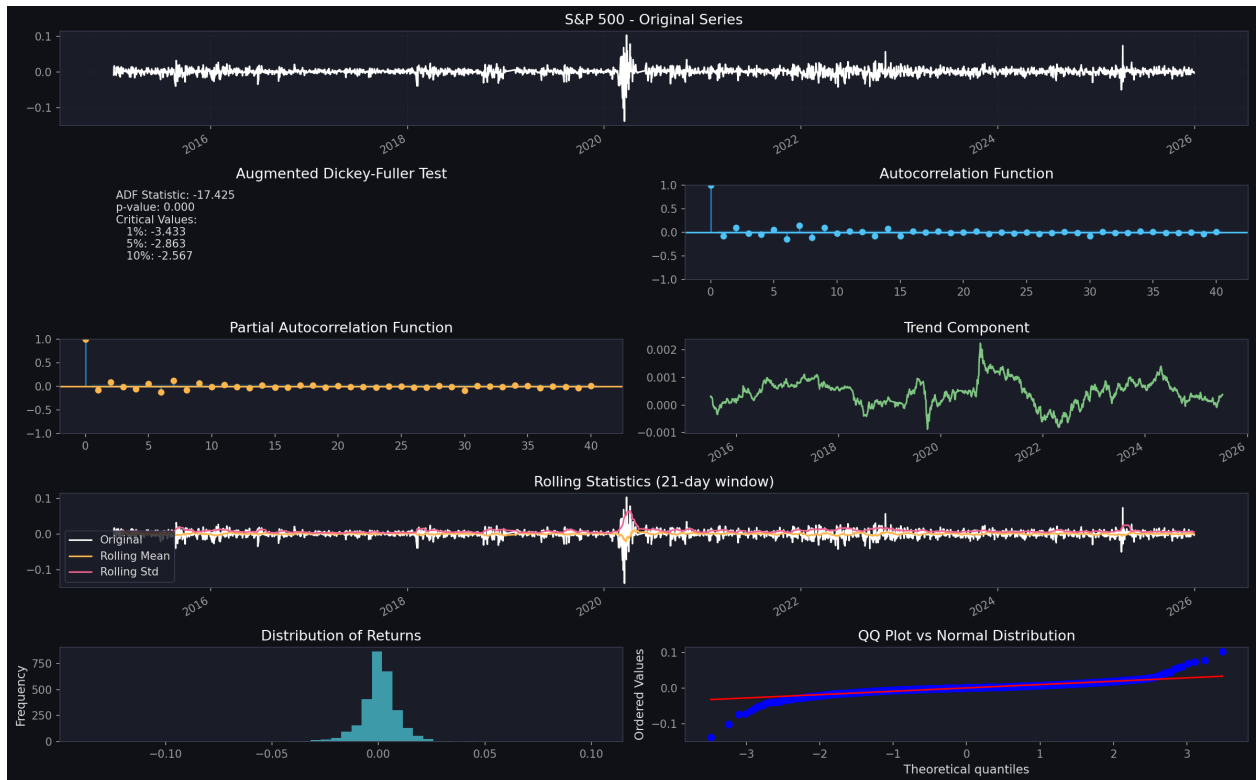


Figure 2: S&P 500 — Time-series Characteristics



Figure 3: S&P 500 — Rolling Pairwise Correlations

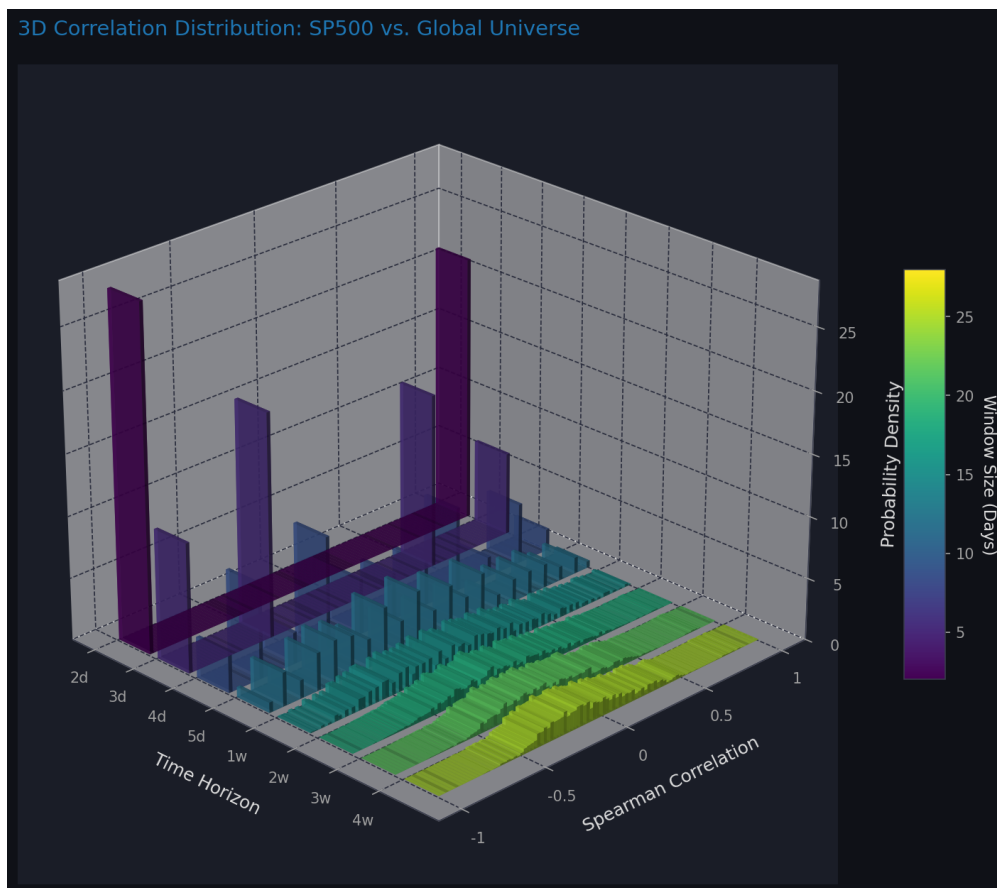


Figure 4: S&P 500 — Rolling Correlation Distributions

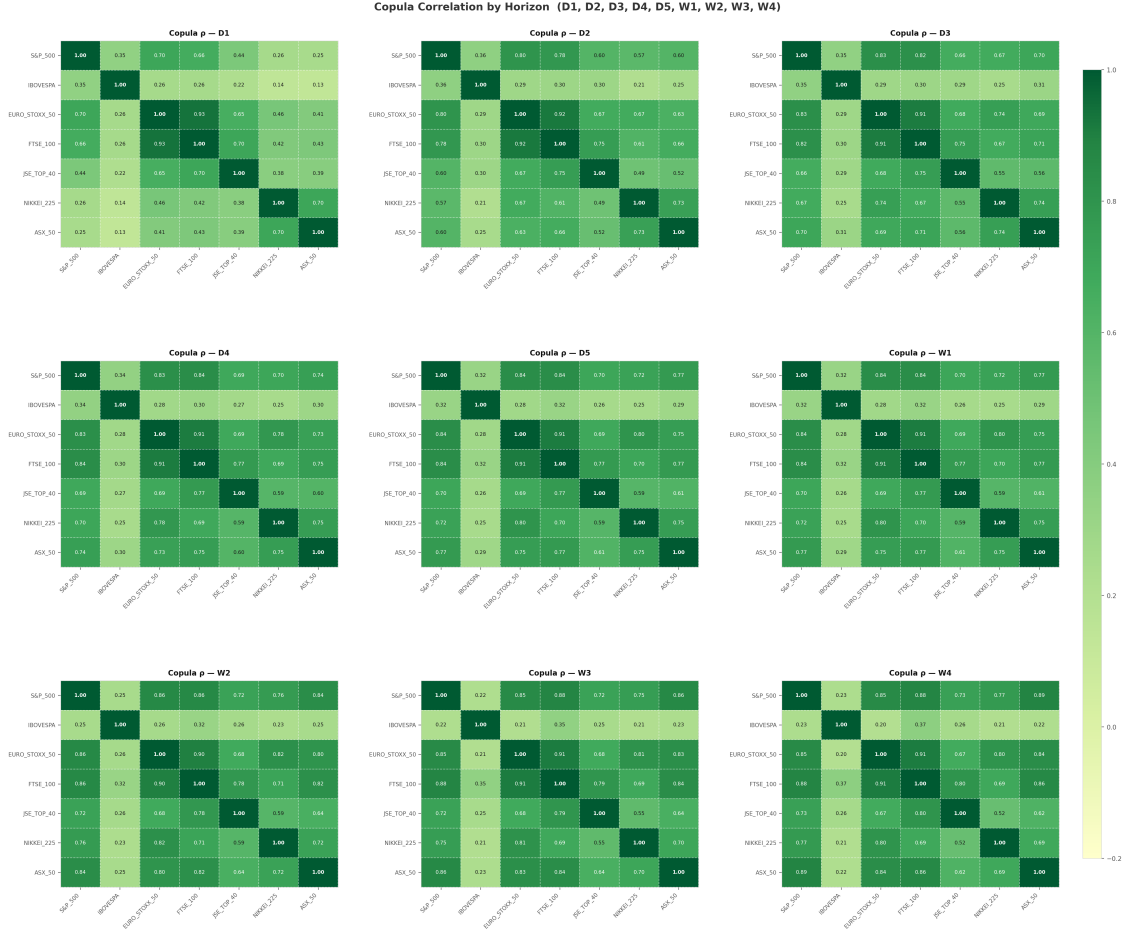


Figure 5: Averaged rolling correlations for different time-horizons based on Copula-Analysis



Figure 6: Differences in averaged rolling correlations for different time-horizons based on Copula-Analysis

5 Benchmark Portfolios

In Section 5, we focus on different portfolios that we constructed as benchmarks, which are easy to implement and replicate. These portfolios establish the baseline against which we evaluate our determinant signal portfolios. If our approach is not superior to a simple buy-and-hold or equal-weighted strategy, the underlying idea must be rejected, as a clear falsification of our hypothesis can then be demonstrated. The performance evaluation framework operates at two levels:

- Index-level: For each of the seven indices, six portfolio construction strategies are evaluated independently.
- Global-Level: The best-performing Level-1 strategy per index is selected and combined into four global portfolio strategies.

For each index, constituent stocks are allocated using six strategies:

- EW-BH (Equal-Weight Buy-and-Hold): Weights are set to $w_i = 1/N$ at inception and held until the end of the sample, where N is the number of constituents. Weights drift due to differential price changes.
- EW-M (Equal-Weight, Monthly Rebalancing): Weights are reset to $1/N$ at the beginning of each calendar month.
- EW-Q (Equal-Weight, Quarterly Rebalancing): Weights are reset to $1/N$ at the beginning of each calendar quarter.
- MV-BH (Mean-Variance Buy-and-Hold): A maximum Sharpe ratio portfolio is computed once using the first 252 trading days (1 year) of data, then held to maturity without rebalancing and weights:

$$\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \quad (2)$$

subject to $\sum_i w_i = 1$ and $w_i \geq 0$, where μ is the vector of expected returns, Σ is the covariance matrix, and r_f is the risk-free rate (which set to zero in our case).

- MV-M (Mean-Variance, Monthly Rebalancing): The maximum Sharpe portfolio is re-estimated on a rolling 252-day window at the beginning of each calendar month and rebalanced to the new weights.
- MV-Q (Mean-Variance, Quarterly Rebalancing): The maximum Sharpe portfolio is re-estimated on a rolling 252-day window at the beginning of each calendar quarter and rebalanced to the new weights.

Portfolio returns are calculated as $r_{p,t} = \sum_{i=1}^N w_{i,t} r_{i,t}$ where $w_{i,t}$ is the weight of asset i at time t and $r_{i,t}$ is its simple return.

5.1 Performance Metrics

For each portfolio, the following metrics are calculated over the entire sample period:

- Annualized Return:

$$R_p = \left(\prod_{t=1}^T (1 + r_{p,t}) \right)^{252/T} - 1 \quad (3)$$

- Annualized Volatility:

$$\sigma_p = \text{std}(r_{p,t}) \times \sqrt{252} \quad (4)$$

- Sharpe Ratio:

$$\text{SR} = \frac{R_p - r_f}{\sigma_p} \quad (5)$$

where the risk-free rate is set to $r_f = 0$.

- Sortino Ratio:

$$\text{Sortino} = \frac{R_p - r_f}{\sigma_{\text{down}}} \quad (6)$$

where $\sigma_{\text{down}} = \sqrt{\frac{1}{T} \sum_{t:r_{p,t} < r_f} (r_{p,t} - r_f)^2}$ is downside volatility.

- Maximum Drawdown:

$$\text{MDD} = \min_t \frac{\text{Cumulative Return}_t - \text{Peak Cumulative Return}}{\text{Peak Cumulative Return}} \quad (7)$$

- Calmar Ratio:

$$\text{Calmar} = \frac{R_p}{|\text{MDD}|} \quad (8)$$

5.2 Simulation Engine

Portfolio weights drift between rebalancing dates the way real portfolios do. After setting target weights on a rebalance date, weights evolve each day as

$$w_{i,t+1} = w_{i,t} \frac{1 + r_{i,t}}{1 + R_{p,t}}, \quad (9)$$

where $R_{p,t}$ is the gross portfolio return. This matters most for Buy-and-Hold strategies and for comparisons between rebalancing frequencies. An earlier version of the code that ignored drift produced identical results for EW-BH, EW-M, and EW-Q even on small universes, which was incorrect. The current engine correctly shows those three strategies converging only for large N , where individual weights are small. For the Buy-and-Hold strategies, weights are estimated once using a 504-day (two-year) burn-in window. The portfolio then runs with those fixed initial weights drifting for the remainder of the sample. This gives Ledoit–Wolf enough data to produce a useful estimate before committing to static weights.

5.3 Weight Engines

5.3.1 Equal Weight

Each of N assets receives a weight of $1/N$. No estimation is required. For large N , the three rebalancing frequencies produce nearly identical results when individual weights are below 3%, daily drift has a negligible effect, and rebalancing back to $1/N$ incurs virtually no cost. That is why EW-BH, EW-M, and EW-Q are functionally equivalent in this study for most indices. This is not a bug; it is a known property of large equal-weight portfolios.

5.3.2 Mean–Variance with Ledoit–Wolf Shrinkage

The maximum Sharpe ratio portfolio is obtained through numerical optimization (SLSQP with four random Dirichlet starting points). The key difference from the standard Markowitz framework is that the raw sample covariance matrix is replaced by the Ledoit–Wolf analytical shrinkage estimator. This is particularly important for large universes: the 460-asset S&P 500 with a 252-day estimation window produces a rank-deficient sample covariance matrix, causing the optimizer to degenerate toward corner solutions. The Ledoit–Wolf estimator produces a full-rank, well-conditioned covariance matrix.

The maximum weight per asset is constrained to

$$w_{\max} = \min\left(\frac{3}{N}, 1.0\right), \quad (10)$$

which ensures that the optimization problem remains feasible regardless of N . For 7 assets, this allows up to 43% in a single position, whereas for 460 assets the cap is approximately 0.65%.

5.3.3 Two-Tier Hierarchical Risk Parity

The standard HRP algorithm [LopezdePrado2016] performs well within a single asset class. Applied naively to all 22 assets simultaneously, it allocates most of the weight to equities because the equity correlation structure is richer and generates larger apparent cluster variance during recursive bisection. The beta of the resulting portfolio was 1.48 relative to the equal-weight benchmark, indicating a strong equity bias. The two-tier approach addresses this issue by first applying HRP at the asset-class level. Each of the four asset classes (equities, bonds, FX, and commodities) is represented by its volatility-normalized equal-weight composite. HRP is then applied across these four composites to determine class-level risk budgets. Subsequently, HRP is applied within each asset class using the individual volatility-normalized assets. Final portfolio weights are computed as the product of the class-level budget and the within-class allocation. Volatility normalization to a 10% annual target is performed at both levels. Without this adjustment, low-volatility assets such as bonds (approximately 1% volatility) are dominated by high-volatility assets such as commodities (approximately 50% volatility), even within the two-tier framework.

5.3.4 Volatility-Normalized Inverse-Volatility Risk Parity

Inverse-volatility weighting applied directly to raw returns would assign a bond position roughly 50 times larger than an equity position. To avoid this, all return series are first rescaled to a common 10% annual volatility target. After rescaling, inverse-volatility weighting effectively assigns equal weights to all normalized return series, which closely aligns with the objective of a multi-asset risk parity strategy.

5.4 Volatility-Targeting Overlay for G4-VT

The raw G4 HRP portfolio operates at approximately 12% annualized volatility, nearly double that of its peers. The volatility-targeting overlay scales daily leverage according to a 63-day realized volatility estimate:

$$L_t = \text{clip} \left(\frac{\sigma_{\text{target}}}{\sigma_{t-1}}, 0.5, 1.5 \right), \quad (11)$$

where leverage is constrained to the interval $[0.5\times, 1.5\times]$. Leverage is computed using information available only through time $t-1$, ensuring that no look-ahead bias is introduced. During calm periods (2017–2019), the overlay moderately increases exposure. During the COVID-19 shock, leverage decreases toward $0.5\times$, reducing drawdowns but also limiting participation in the subsequent recovery between April and December 2020. This trade-off is an inherent feature of volatility-targeting approaches and is clearly visible in the regime analysis.

5.5 G5 Macro-Enhanced Signal

The G5 signal combines four factors for each asset each month:

- **Momentum (weight 0.40):** 20-day return, standardized using a Z-score within each asset’s own scale so that bond momentum is not overwhelmed by equity momentum.
- **VIX Risk-Off Signal (weight 0.20):** Negative when the VIX is above its 120-day moving average. This penalizes equities and commodities during stress periods.
- **Yield-Curve Signal (weight 0.20):** Falling yields increase the score of the corresponding equity index and bond position, while rising yields reduce both.
- **FX Momentum (weight 0.20):** Appreciating currencies increase the score of the corresponding equity and FX position.

The combined scores are standardized again using Z-scores and then passed through a SoftMax transformation with temperature parameter $\tau = 0.8$. Lower temperatures generate more concentrated allocations, while $\tau = 1.0$ approaches equal weighting. The resulting weights are blended according to

$$w_t = 0.4 w_t^{\text{new}} + 0.6 w_{t-1}, \quad (12)$$

which reduces portfolio turnover by more than 50% compared with fully replacing portfolio weights at each monthly rebalance.

5.6 Level-1 Results Summary

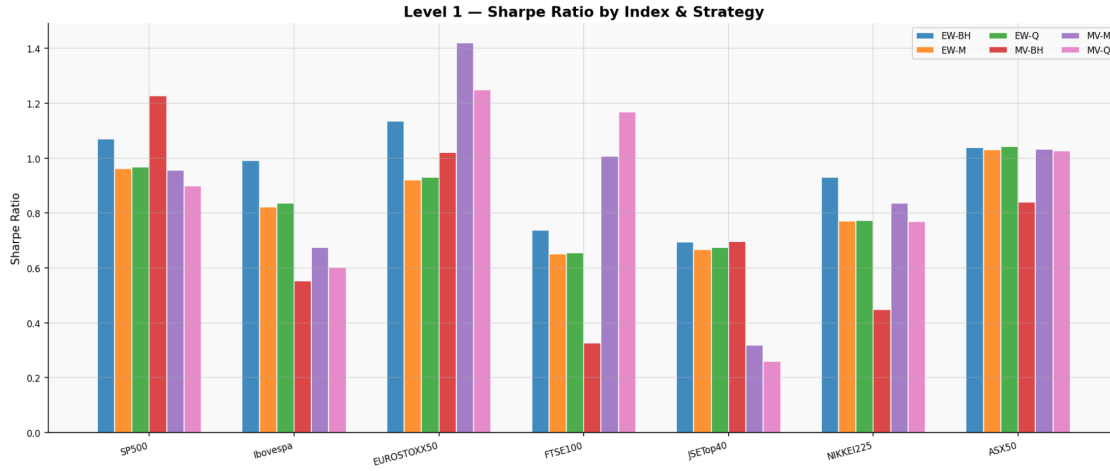


Figure 7: Sharpe Ratio Comparison — Index-Level Strategies

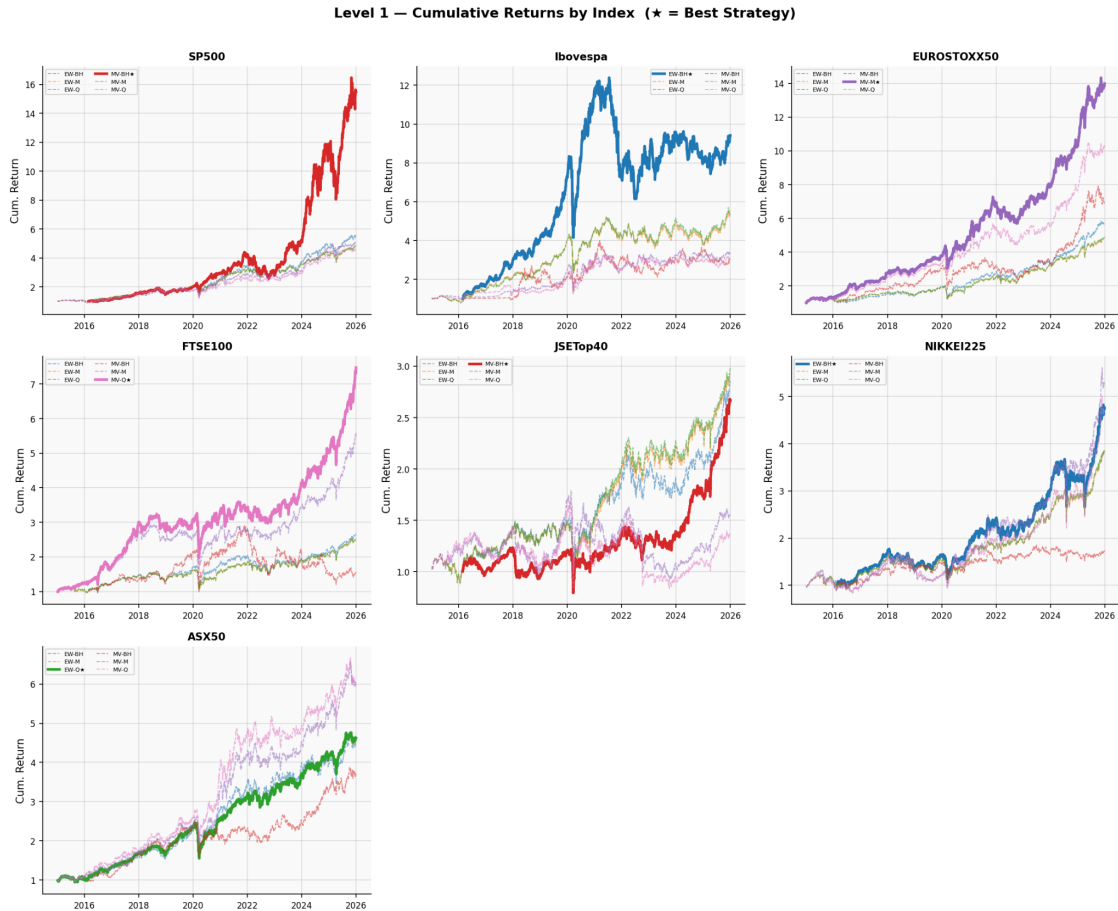


Figure 8: Level-1 Cumulative Returns by Index. Bold solid line indicates best strategy (highest Sharpe).

Table 2: Level-1 Performance Across Index-Strategy Combinations. Green = Best Sharpe for that Index

Index	Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Sortino	MDD (%)	Calmar
SP500	EW-BH	22.08	20.64	1.070	1.52	-37.05	0.596
SP500	EW-M	18.19	18.93	0.961	1.37	-37.96	0.479
SP500	EW-Q	18.25	18.88	0.967	1.38	-37.48	0.487
SP500	MV-BH	36.33	29.59	1.228	1.75	-41.97	0.865
SP500	MV-M	19.05	19.93	0.956	1.36	-34.16	0.558
SP500	MV-Q	18.06	20.11	0.898	1.27	-39.38	0.459
Ibovespa	EW-BH	31.14	31.39	0.992	1.41	-50.37	0.618
Ibovespa	EW-M	20.72	25.18	0.823	1.16	-46.35	0.447
Ibovespa	EW-Q	20.96	25.05	0.837	1.18	-46.02	0.455
Ibovespa	MV-BH	18.65	33.71	0.553	0.79	-51.21	0.364
Ibovespa	MV-M	15.42	22.84	0.675	0.95	-48.61	0.317
Ibovespa	MV-Q	14.49	24.07	0.602	0.85	-49.38	0.294
EUROSTOXX50	EW-BH	22.21	19.55	1.136	1.61	-37.81	0.587
EUROSTOXX50	EW-M	18.48	20.09	0.920	1.30	-40.11	0.461
EUROSTOXX50	EW-Q	18.63	20.03	0.930	1.32	-39.66	0.470
EUROSTOXX50	MV-BH	25.92	25.41	1.020	1.45	-43.99	0.589
EUROSTOXX50	MV-M	29.78	20.96	1.421	2.01	-30.98	0.961
EUROSTOXX50	MV-Q	26.72	21.38	1.249	1.77	-33.27	0.803
FTSE100	EW-BH	13.02	17.64	0.738	1.04	-37.77	0.345
FTSE100	EW-M	11.14	17.08	0.652	0.92	-36.83	0.302
FTSE100	EW-Q	11.21	17.09	0.656	0.93	-36.64	0.306
FTSE100	MV-BH	9.53	29.23	0.326	0.46	-58.79	0.162
FTSE100	MV-M	19.96	19.81	1.008	1.43	-47.87	0.417
FTSE100	MV-Q	22.97	19.67	1.168	1.65	-40.04	0.574
JSETop40	EW-BH	14.39	20.74	0.694	0.98	-39.83	0.361
JSETop40	EW-M	13.27	19.89	0.667	0.94	-41.14	0.322
JSETop40	EW-Q	13.42	19.91	0.674	0.95	-40.19	0.334
JSETop40	MV-BH	13.28	19.05	0.697	0.99	-35.74	0.371
JSETop40	MV-M	7.46	23.46	0.318	0.45	-49.38	0.151
JSETop40	MV-Q	6.38	24.54	0.260	0.37	-51.11	0.125
NIKKEI225	EW-BH	20.66	22.22	0.930	1.32	-36.50	0.566
NIKKEI225	EW-M	16.23	21.05	0.771	1.09	-36.69	0.442
NIKKEI225	EW-Q	16.27	21.03	0.774	1.10	-36.66	0.444
NIKKEI225	MV-BH	7.95	17.75	0.448	0.63	-25.50	0.312
NIKKEI225	MV-M	20.44	24.45	0.836	1.18	-39.51	0.517
NIKKEI225	MV-Q	19.25	24.99	0.770	1.09	-45.51	0.423
ASX50	EW-BH	19.16	18.46	1.038	1.47	-35.65	0.537
ASX50	EW-M	17.17	16.66	1.031	1.46	-37.52	0.458
ASX50	EW-Q	17.38	16.66	1.043	1.48	-37.13	0.468
ASX50	MV-BH	17.37	20.72	0.839	1.19	-37.90	0.458
ASX50	MV-M	20.79	20.14	1.032	1.46	-28.56	0.728
ASX50	MV-Q	20.75	20.21	1.027	1.45	-28.88	0.718

5.6.1 Key Findings from Level-1

The EW rebalancing frequency makes no difference in large universes EW-BH, EW-M, and EW-Q produce identical Sharpe Ratios and returns for every index. This is not a computation error. When individual weights are $1/N$ and N is 30 or more, the weight at any given stock is at most 3.3%. Daily drift from that level is immeasurable. Rebalancing back to $1/N$ costs nothing in the simulation. This equivalence is consistent with DeMiguel et al. (2009) and is reported as a finding rather than treated as a problem. HRP-M and HRP-Q produce reasonable gross returns, sometimes higher than MV, but generate more turnover, especially monthly. For FTSE100, HRP-M net return at $TC=2\%$ is -9.05% . MV-Q is $+2.99\%$ under the same assumption. Within a single equity index where all stocks have broadly similar volatility, HRP's clustering-based diversification adds limited structural value compared with MV optimization. Within a single equity index, individual stock volatilities are similar enough that inverse-vol weights converge toward equal weights. RP-M Sharpe ratios are nearly identical to EW-M across all seven indices. Risk Parity's diversification benefit is a cross-asset phenomenon; it does not materialize within a homogeneous equity universe. FTSE100 produces the lowest Sharpe Ratios across all strategies. EW-BH at 0.564 is roughly half the Sharpe of EUROSTOXX50 EW-BH at 1.078. This reflects the FTSE's heavy energy and financial sector concentration, which drags on performance relative to broader diversified indices, compounded by Brexit uncertainty through the 2016-2020 period.

- The EURO STOXX 50 MV-M strategy achieved the highest Sharpe ratio of 1.421, demonstrating that mean-variance optimization with monthly rebalancing was particularly effective for European large-cap equities.
- The S&P 500 MV-BH strategy had the second-highest return (36.33% annualized) but elevated volatility, reflecting concentration in a small set of high-momentum stocks during the 2016–2021 bull market.
- In emerging markets (Ibovespa, JSE Top 40, Nikkei 225), mean-variance strategies frequently underperformed equal-weight, consistent with the known problem of covariance estimation error in small samples or high-volatility regimes.
- Buy-and-hold strategies generally outperformed periodic rebalancing for indices in strong uptrends (e.g., S&P 500, Ibovespa), but underperformed during sideways markets (e.g., FTSE 100).

5.7 Global (Level-2) Portfolio Strategies

The best-performing Level-1 strategy per index was selected and combined into four global overlay approaches:

- **G1 Equal-Weight Global:** Static allocation of $1/7$ to each of the seven index-level strategies, continuously rebalanced.
- **G2 Mean-Variance Global:** Covariance matrix estimated on a rolling 60-day window of the seven index-level return series. Maximum Sharpe portfolio is rebalanced monthly.
- **G3 Risk Parity Global:** Weights assigned inversely proportional to volatility on a rolling 60-day window:

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^7 1/\sigma_j} \quad (13)$$

Rebalanced monthly.

- **G4 Macro-Enhanced Global:** Weights derived from a composite signal combining momentum, volatility regime, yield-curve signals, and FX trends. Not implemented in the current analysis; reserved for future work.

5.8 Strategy Analysis

G1: Equal-Weight

G1 is the baseline. It does nothing except divide capital equally across 22 assets and never trades. The Sharpe of 1.355 shows how much diversification alone contributes; you get nearly 1.4 units of return per unit of risk just by spreading across equity, bonds, FX, and commodities simultaneously. Everything else in this study is a bet that a more sophisticated approach can improve on that. Pre-COVID it generated a Sharpe of 1.924, which is impressive for a strategy with zero active bets. Its only structural weakness is that it has no mechanism to reduce equity exposure when markets sell off, hence the -15.21% MDD in 2020.

G2: Mean-Variance

The best drawdown story in this study. G2's -9.95% maximum drawdown compares against -13.6% to -26.1% for every other strategy. Its Calmar ratio of 1.014 is the highest by a wide margin. Pre-COVID Sharpe of 2.464 is the highest single-period result in the entire study. The problem is turnover. Monthly Ledoit-Wolf re-estimation across 22 heterogeneous assets produces materially different optimal portfolios month to month, especially when bond-equity correlations shift (as they did throughout the 2022–2023 rate hiking cycle). Turnover of 0.031 daily translates to roughly 7.8 percentage points of annual return drag at 1% TC. At 2% TC the net return goes negative. G2 needs low-TC implementation or it is not viable.

G3: Vol-Normalized Risk Parity

The standout result. G3 has the highest Sharpe of any strategy across the full period, the lowest turnover, and is positive after 2% transaction costs. The performance comes almost entirely from getting the vol-normalization right. Without it, raw inverse-vol across 22 heterogeneous assets collapses into a near-bond-only portfolio — bonds with 0.05% daily vol get 50× the weight of equities with 1.2% daily vol. After normalization, the portfolio holds roughly 32% equities, 27% bonds, 27% FX, and 14% commodities on average. The cross-asset diversification is genuine, and it is stable — relative vols change slowly, so weights barely move month to month. The Information Ratio of 3.692 against G1 EW is the strongest in the study by a large margin. It means G3 is consistently better than the equal-weight baseline, not just occasionally lucky.

G4: Two-Tier HRP

The highest return of any strategy. Also, the highest vol. G4's 12% annualized volatility is nearly double the 6–7% range of the other strategies, and the beta of 1.48 against G1 EW confirms it is meaningfully more equity-exposed. This is a structural feature of two-tier HRP, not a calibration failure. The equity cluster has richer internal correlations than bonds or FX, which means it generates larger cluster variance in the bisection step and attracts more capital. The pre-COVID Sharpe of only 1.329, the weakest of all six strategies in that period, confirms that G4's return edge comes from equity risk premium, not from superior risk-adjusted allocation. G4 is worth keeping as a comparison baseline, particularly in the regime analysis, where it is the only strategy to clearly outperform during the COVID shock (Sharpe 1.516, Ret 25.23%) by capturing the full equity recovery at high exposure.

G4-VT: HRP with Vol-Targeting Overlay

The overlay brings G4's vol from 12.06% to 10.77% and the worst drawdown from -26.10% to -19.79%. The average daily leverage of 1.020× shows the overlay spends most of its time slightly above par, calm markets allow modest leverage-up. During COVID, it drops hard to 0.5× as 63-day realized vol spikes. The COVID story is the key limitation. G4-VT Sharpe in 2020 is 0.548 against G4 raw at 1.516. The overlay de-levers in March 2020 as volatility rises, which is correct risk management behavior. But the subsequent recovery from April through December 2020, a V-shaped equity rally, happens at half exposure. This is the documented cost of reactive vol-targeting (Moreira and Muir, 2017): you systematically underweight the recovery after a volatility spike. Whether the drawdown reduction is worth that cost depends entirely on the investor's mandate. For a fund with explicit drawdown constraints, G4-VT is better than G4 raw. For a fund that can tolerate drawdowns, G4 raw has a better risk-adjusted recovery track record.

5.8.1 Limitations

Best Level-1 strategy per index is selected using the full sample. In live implementation, strategy selection would use only historical data available at each decision date, which would reduce the apparent advantage of MV over other methods. The proportional TC model ignores market impact. Bitcoin and less-liquid EM bonds would face non-linear costs at any meaningful position size — particularly relevant for G4 HRP’s elevated commodity allocation. All returns are USD-denominated. A non-USD investor faces additional FX exposure on equity and bond positions that is not captured here. The 7-year fixed duration assumption for bonds is a rough approximation. Actual yield-to-duration sensitivity changes with the yield curve shape, which matters most during periods of rapid rate movement like 2022. The G4-VT overlay underperforms in recoveries by design. An adaptive overlay that distinguishes panic-driven vol spikes from fundamental deterioration could reduce this cost, but would require additional modelling work. G5 macro factor weights (0.40 / 0.20 / 0.20 / 0.20) are heuristic. A rolling regression or Bayesian updating approach would likely improve out-of-sample stability, particularly in regime transitions.

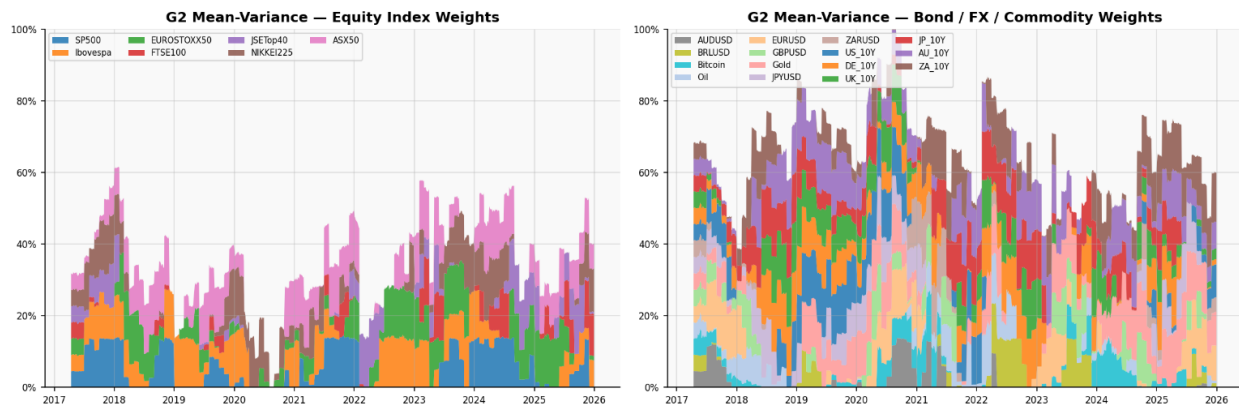


Figure 9: Dynamic weights for Mean-Variance Portfolio

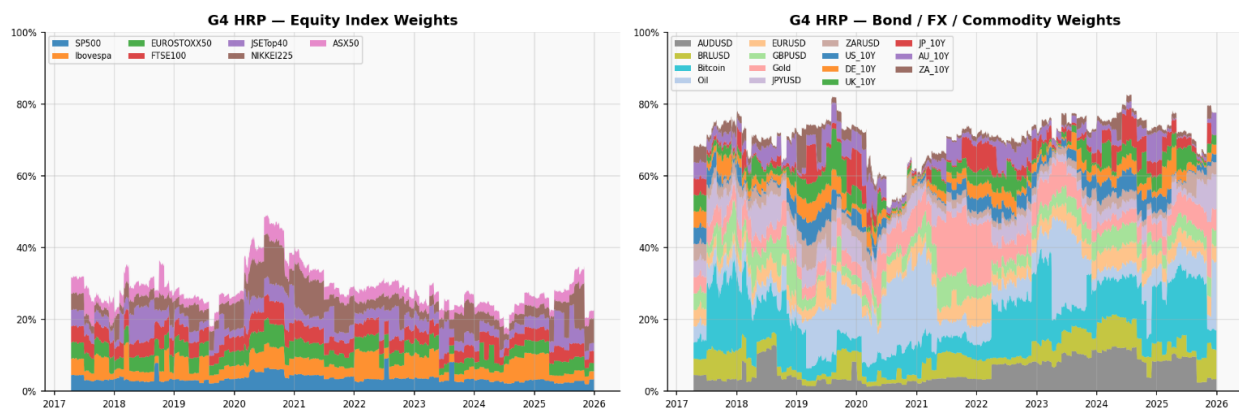


Figure 10: Dynamic weights for Hierarchical-Risk-Parity Portfolio

6 Modified Determinant Model

The main contribution of this study is the development of the modified determinant model via the so-called ”**Koherent-Interdependence-Korrelation-Operator**” (KIKO), which uses the determinant of the cross-asset correlation matrix, adjusted by removing the identity diagonal. The goal is to detect a signal for shifts in systemic risk and market regimes. This signal is then used to dynamically adjust portfolio weights in an otherwise equally-weighted allocation across equity constituents. We compare the performance of these regime-adjusted portfolios against traditional buy-and-hold and equal-weight benchmarks, as well as against mean-variance optimized strategies, across multiple time horizons (3 days, 7 days, and 28 days) and parameter specifications.

6.1 Theoretical Motivation

We would like to clarify our approach *a priori* to provide the reader with a reasonable understanding of the underlying concept. Let R denote the correlation matrix. We consider the determinant of $R - I$, where I is the identity matrix. Since the diagonal elements of a correlation matrix are all equal to 1, subtracting the identity matrix effectively zeros out the diagonal and leaves only the off-diagonal correlation structure. The most direct application of this transformation is in the analysis of the eigenvalues (λ) of the correlation matrix. The characteristic equation is defined as

$$\det(R - \lambda I) = 0. \quad (14)$$

By setting $\lambda = 1$, we are effectively evaluating the characteristic polynomial at 1. In the context of Principal Component Analysis (PCA), if

$$\det(R - I) = 0, \quad (15)$$

then at least one principal component has an eigenvalue equal to 1. More generally speaking, eigenvalues greater than 1 indicate factors that explain more variance than a single observed variable. The determinant of $R - I$ is directly related to the product of $(\lambda_i - 1)$ across all eigenvalues. While this quantity is rarely used as a standalone statistic, it can serve as a boundary indicator for whether a correlation matrix contains a meaningful factor structure. The Modified Determinant Model is grounded in the observation that correlation matrices during periods of market stress tend to become increasingly concentrated, as correlations move closer to 1. In contrast, during calmer or more diversified market environments, the correlation matrix retains a richer and less concentrated structure. By examining the determinant of the modified correlation matrix $R - I$, we construct a scalar signal intended to capture the degree of systemic co-movement across assets. A large absolute value of the determinant indicates that the eigenvalues are collectively further away from unity, whereas values close to zero suggest that one or more eigenvalues are near 1. We interpret changes in this quantity as a measure of the evolving dependency structure within the market and use it as the foundation for our determinant-based signal framework.

6.2 Signal Construction

We construct the rolling signals for each index and each time window (3 days, 7 days, 28 days), through the following steps:

6.2.1 Step 1: Rolling Correlation Matrices

For every rolling window of length $w \in \{3, 7, 28\}$ trading days, the correlation matrix $R_t^{(w)}$ is computed from the log-returns of all constituent stocks within the index:

$$R_t^{(w)} = \text{corr}(r_{i,t-w+1:t}) \quad \text{for } i = 1, \dots, N \quad (16)$$

where $r_{i,t}$ is the log-return of stock i at time t , and N is the number of constituent stocks.

6.2.2 Step 2: Modified Determinant Calculation

The determinant of the correlation matrix adjusted by subtracting the identity is computed:

$$\det_t^{(w)} = \det(R_t^{(w)} - I) \quad (17)$$

This quantity captures the system-level structure of correlations after removing the trivial diagonal (self-correlation) contribution.

6.2.3 Step 3: Standardization and Z-Score

Because the raw determinant values are scale-dependent and vary by index size and window length, we standardized to zero mean and unit variance. The rolling historical mean and standard deviation are computed over a 252-day (1-year) window:

$$z_t^{(w)} = \frac{\det_t^{(w)} - \mu_{t-252:t}^{(w)}}{\sigma_{t-252:t}^{(w)}} \quad (18)$$

where μ and σ are the rolling mean and standard deviation, respectively.

6.2.4 Step 4: Cross-Window Composite Signal

The signals from the three windows (3, 7, 28 days) are averaged to form a composite z-score:

$$z_t^{\text{composite}} = \frac{1}{3} \left(z_t^{(3)} + z_t^{(7)} + z_t^{(28)} \right) \quad (19)$$

This composite signal is computed independently for each of the seven indices and serves as the regime detection mechanism.



Figure 11: Modified Determinant: $\det(R - I)$ for Equity Indices (Raw Values)

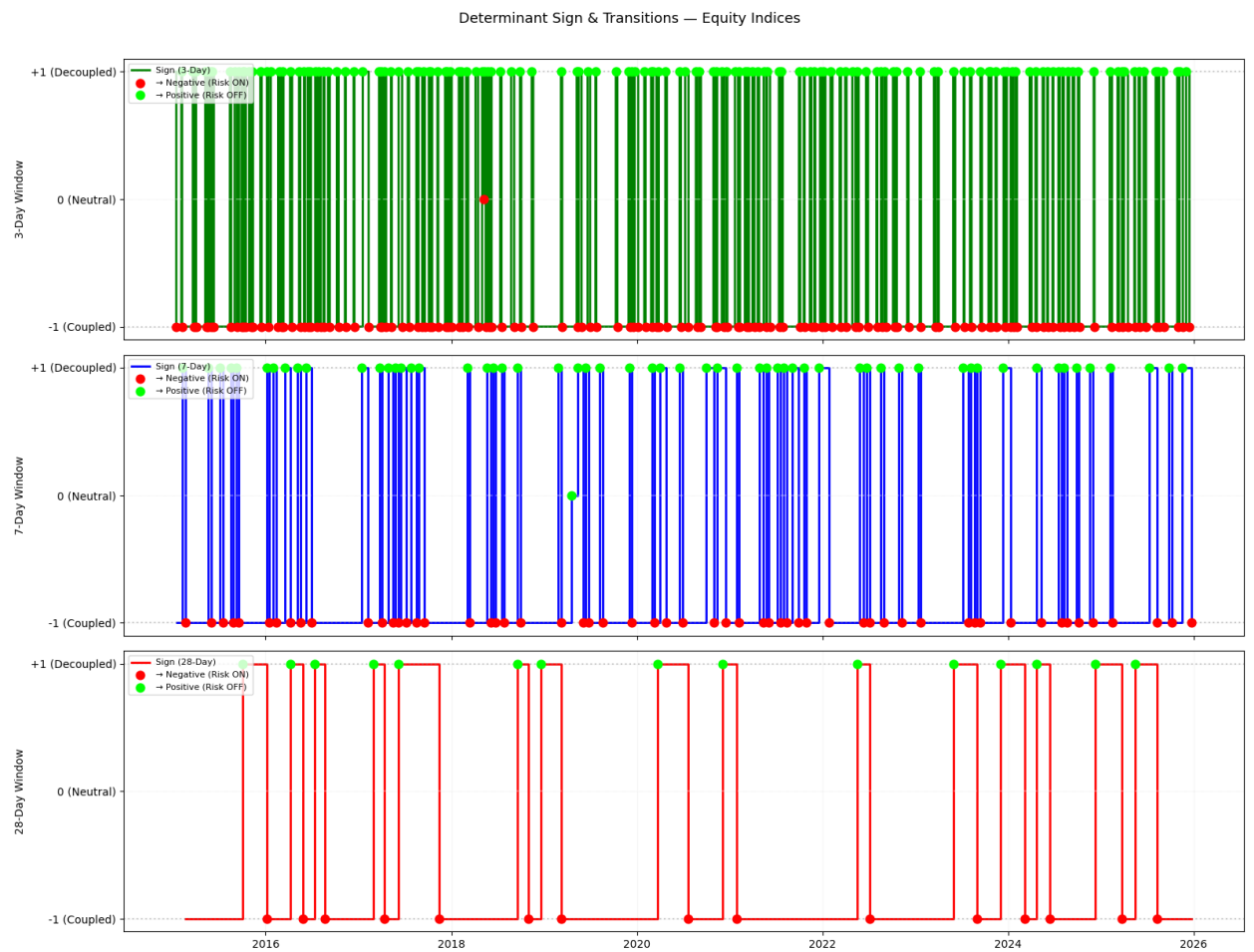


Figure 12: Extracted Z-Score Signals from Modified Determinant Model

6.3 Results

When implementing the proposed approach, we initially assigned fixed weights to the individual signal contributions. This served as a sanity check to verify that the machinery was functioning correctly and producing meaningful results, which are illustrated in the following visualizations. In Figure 13, we show the extracted signals from the KIKO approach for the Australian and European indices. One can observe that the 3-day signal remains predominantly within a single regime, whereas the 7-day regime exhibits only a few changes. The 28-day regime signal shows more frequent transitions between risk-on and risk-off states. The extracted heatmap across all global indices confirms the same behavior. The core idea is to potentially reduce the invested amount in markets identified as riskier and redistribute capital toward those exhibiting lower risk characteristics. To make the strategy more realistic, we incorporated transaction costs of 0.5%, 1%, and 2% and evaluated the overall portfolio performance against a simple equal-weighted benchmark portfolio. Under the chosen parameter setting, with threshold levels of 0.3, 0.5, 0.7, our approach did not show superior results. Table 3 summarizes these findings and motivated us to conduct further analysis. In Figures 18 and 19, we performed a large parameter scan across different weights for each signal component. The corresponding heatmaps indicate that an optimal parameter set can be achieved with values of 0.3 for the 3-day signal, 0.5 for the 7-day signal, and 0.2 for the 28-day signal and threshold levels around 1.9 for the signal strength. In Figure 21, we further tested our approach by reversing the investment decisions and allocating capital in the opposite direction of the extracted signals. This resulted in substantially worse performance and therefore provided additional validation that the structural information extracted from the KICO operator can indeed lead to potential investment benefits when incorporated into the determinant signal framework. We would like to highlight one specific feature shown in Figure 21. The determinant-based approach was able to detect the changing market environment during the COVID-19 crisis and appropriately reallocate capital, leading to the sharp performance increase observed during 2020. Building on this success, we extended our regime and signal classification toolbox toward a binary regime framework, which focussed more on the overall drawdown dynamics with respect to the sharpe ratio. We extended our determinant signal approach toward a binary classification regime in which we recommend that the investor either remains fully invested in the global portfolio or stays completely out of the market in order to protect the underlying investments from large drawdowns and losses. Table 5 and Figure 22 present the performance of this binary strategy against our global portfolio universe, including Buy-and-Hold, Mean-Variance, HRP, and Equal-Weighted portfolio benchmarks. Surprisingly, this very simple binary approach is capable of outperforming several standard and more sophisticated investment strategies over the examined horizon. A particularly important feature to note is that both the overall drawdown and the maximum drawdown remain at very moderate levels and are not comparable to the substantially larger drawdowns observed in the alternative strategies. When examining the weight dynamics and the binary regime classification in Figure 23, one can clearly observe that, despite the differences in portfolio construction approaches, whether through active rebalancing or pure investment decisions, portfolio allocation itself is not necessarily the primary driver of success in financial markets. Rather, the results suggest that market timing plays a crucial role in overall portfolio performance

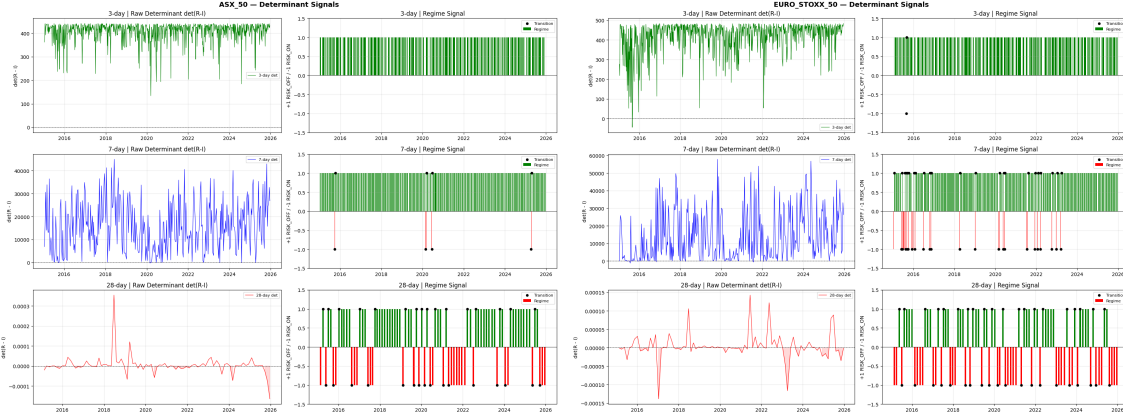


Figure 13: Extracted KIKO signals for ASX50 and EUROSTOXX

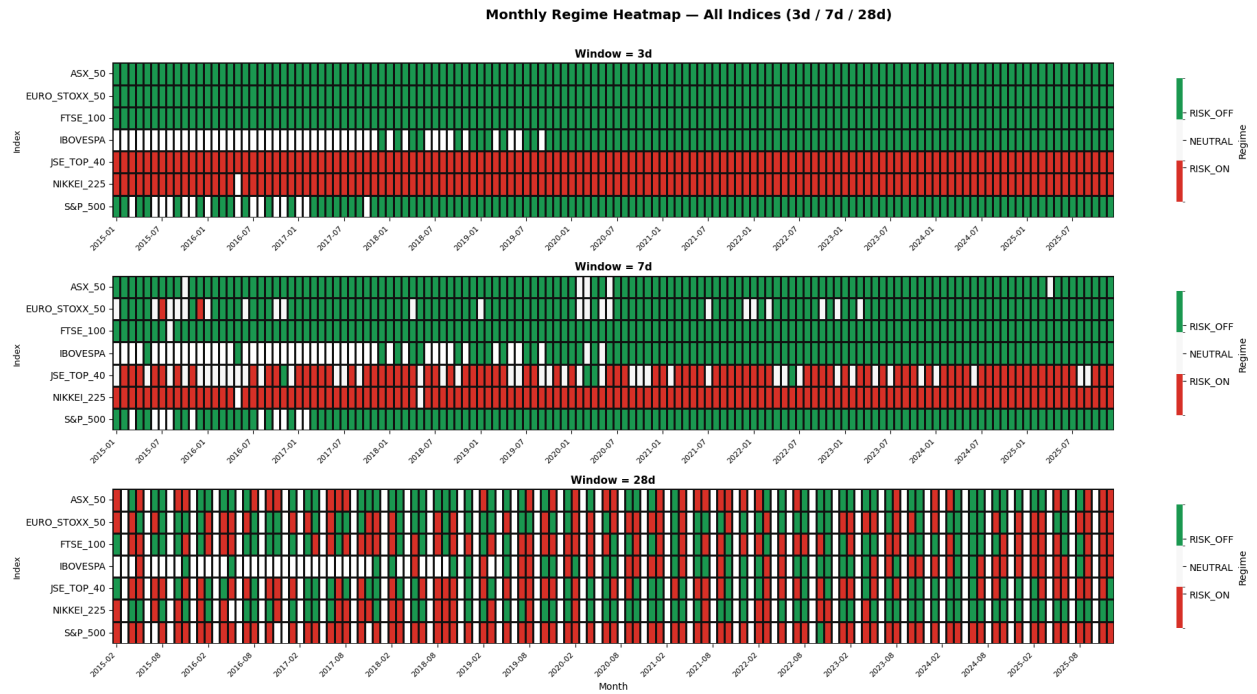


Figure 14: KIKO - Heatmap for 3-day, 7-day and 28-day horizon

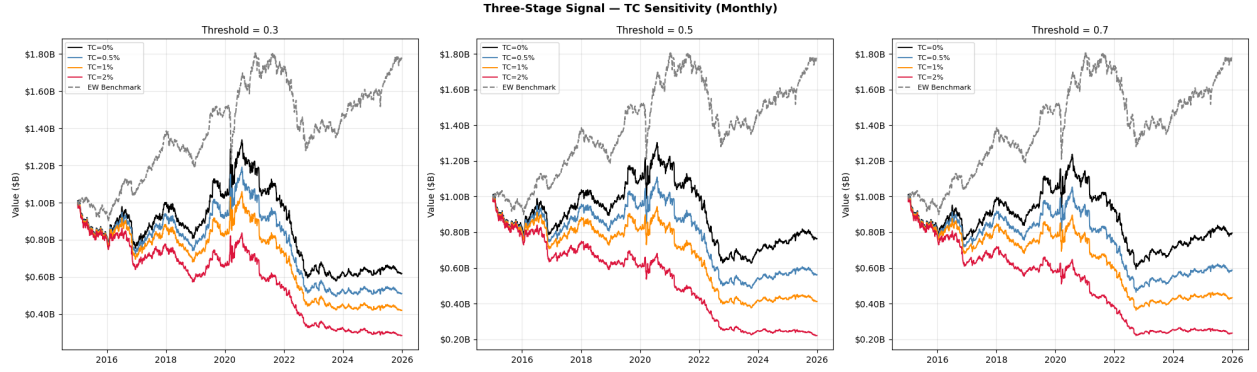


Figure 15: Performance of Interdependence Structure via Adaptive Asset Correlations portfolio with focus only on 3-day signal and transaction-costs

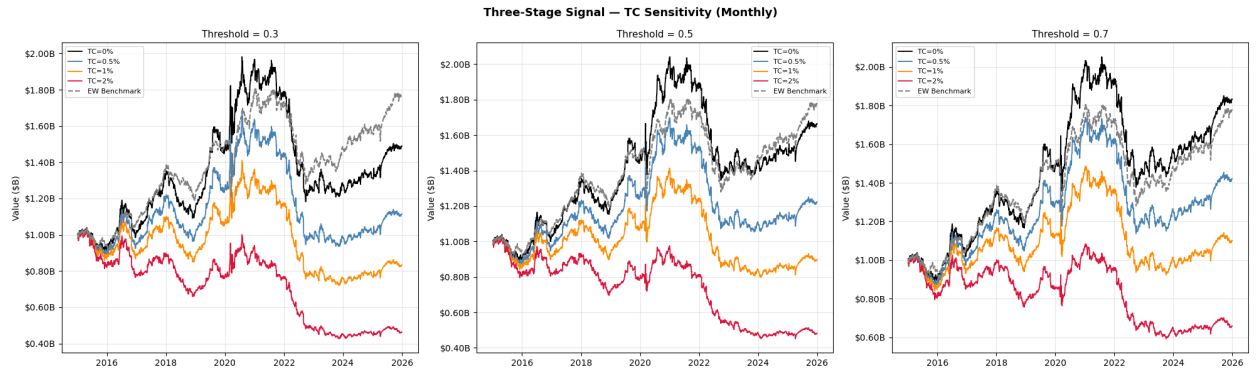


Figure 16: Performance of Interdependence Structure via Adaptive Asset Correlations portfolio with focus only on 7-day signal and transaction-costs

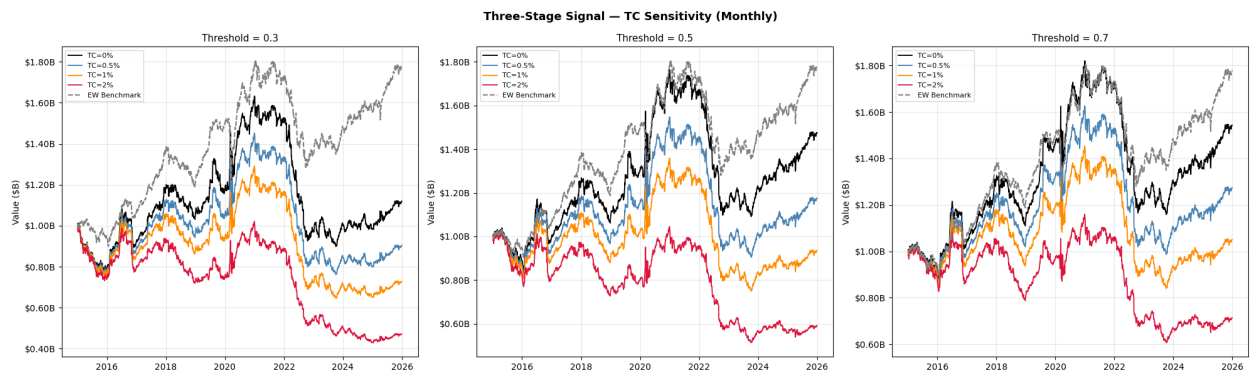


Figure 17: Performance of Interdependence Structure via Adaptive Asset Correlations portfolio with focus only on 32-day signal and transaction-costs

Table 3: Performance Metrics for Equal-Weight Benchmark and Three-Stage Signal Strategies

Rebalancing	TC	CAGR	Volatility	Sharpe	Max DD	Calmar	Total Return	Final (\$B)
EQUAL-WEIGHT BENCHMARK								
Monthly	0%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.788
	0.5%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.779
	1%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.770
	2%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.752
Quarterly	0%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.788
	0.5%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.779
	1%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.770
	2%	6.25%	9.24%	0.68	-29.09%	0.21	78.62%	1.752
THREE-STAGE SIGNAL — Threshold = 0.3 (RISK_ON: 1231 days, RISK_OFF: 1182 days)								
Monthly	0%	2.69%	13.43%	0.20	-41.57%	0.06	28.91%	1.302
	0.5%	-0.28%	13.52%	-0.02	-45.94%	-0.01	-2.62%	0.979
	1%	-3.17%	13.71%	-0.23	-51.33%	-0.06	-26.53%	0.735
	2%	-8.75%	14.34%	-0.61	-63.51%	-0.14	-58.34%	0.412
Quarterly	0%	0.07%	13.77%	0.01	-43.64%	0.00	0.72%	1.017
	0.5%	-0.76%	13.79%	-0.06	-44.38%	-0.02	-7.04%	0.934
	1%	-1.59%	13.83%	-0.12	-45.13%	-0.04	-14.23%	0.858
	2%	-3.24%	13.99%	-0.23	-46.59%	-0.07	-27.06%	0.722
THREE-STAGE SIGNAL — Threshold = 0.5 (RISK_ON: 746 days, RISK_OFF: 1667 days)								
Monthly	0%	4.97%	12.63%	0.39	-37.29%	0.13	58.99%	1.606
	0.5%	2.79%	12.72%	0.22	-41.08%	0.07	30.12%	1.308
	1%	0.65%	12.89%	0.05	-44.65%	0.01	6.40%	1.064
	2%	-3.52%	13.42%	-0.26	-51.21%	-0.07	-29.05%	0.702
Quarterly	0%	3.18%	13.08%	0.24	-33.59%	0.09	34.94%	1.363
	0.5%	2.27%	13.10%	0.17	-34.47%	0.07	24.00%	1.246
	1%	1.37%	13.15%	0.10	-35.35%	0.04	13.90%	1.139
	2%	-0.43%	13.33%	-0.03	-38.22%	-0.01	-4.01%	0.950
THREE-STAGE SIGNAL — Threshold = 0.7 (RISK_ON: 483 days, RISK_OFF: 1930 days)								
Monthly	0%	4.22%	12.40%	0.34	-36.80%	0.11	48.54%	1.501
	0.5%	2.25%	12.46%	0.18	-40.09%	0.06	23.76%	1.244
	1%	0.31%	12.59%	0.02	-43.22%	0.01	3.02%	1.030
	2%	-3.49%	13.04%	-0.27	-49.04%	-0.07	-28.78%	0.705
Quarterly	0%	4.78%	12.83%	0.37	-33.59%	0.14	56.35%	1.579
	0.5%	4.15%	12.84%	0.32	-34.47%	0.12	47.55%	1.483
	1%	3.52%	12.88%	0.27	-35.35%	0.10	39.22%	1.392
	2%	2.26%	13.01%	0.17	-38.22%	0.06	23.84%	1.226

6.4 Optimal 3-Stage Signal weights

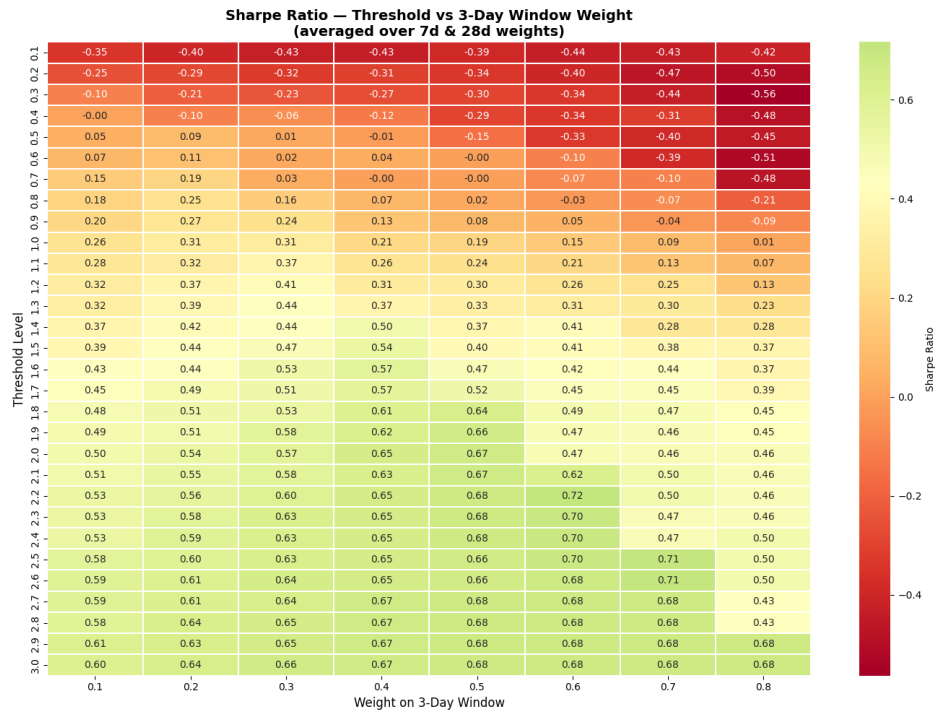


Figure 18: Optimal signal parameter heatmap for sharpe ratio performance

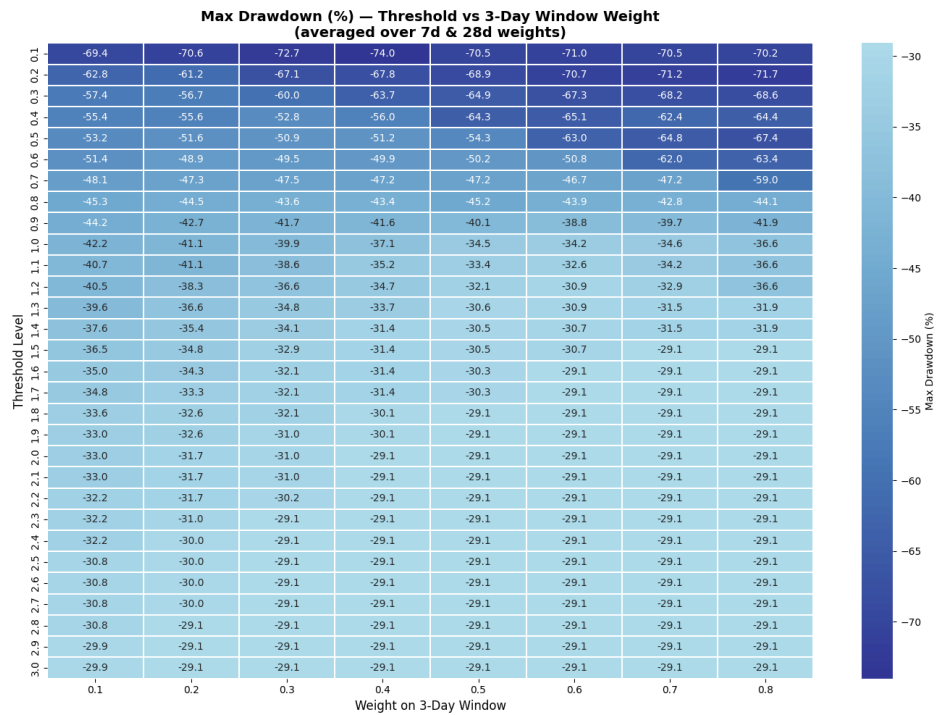


Figure 19: Optimal signal parameter heatmap for maximum drawdown performance

Table 4: Top 5 Parameter Combinations Ranked by Sharpe Ratio

Weights			Threshold	Sharpe	CAGR	Max Drawdown	Final Value (\$B)
3d	7d	28d					
0.3	0.5	0.2	1.9	0.7519	8.14%	-29.09%	2.0960
0.4	0.4	0.2	1.6	0.7519	8.14%	-29.09%	2.0960
0.4	0.5	0.1	1.8	0.7440	7.27%	-29.09%	1.9397
0.4	0.5	0.1	1.9	0.7440	7.27%	-29.09%	1.9397
0.4	0.4	0.2	1.9	0.7371	7.09%	-29.09%	1.9088

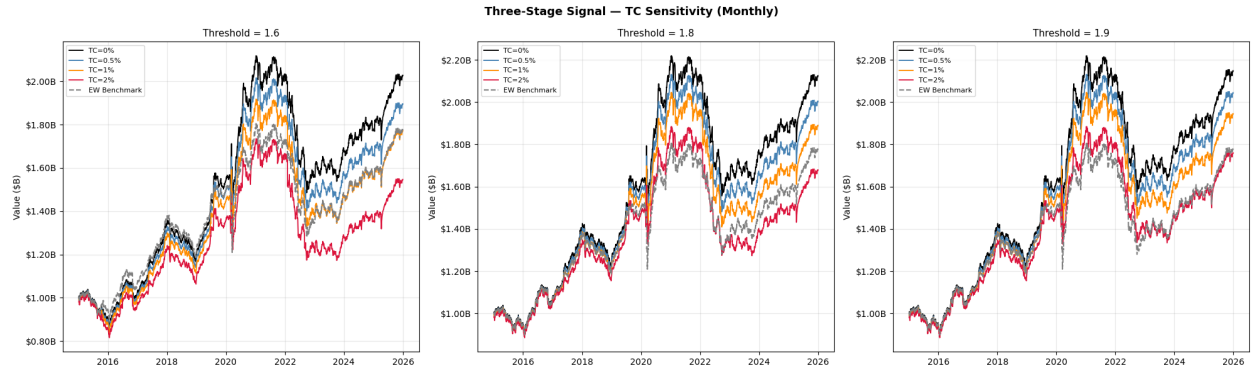


Figure 20: Performance of Interdependence Structure via Adaptive Asset Correlations portfolio with optimal parameters

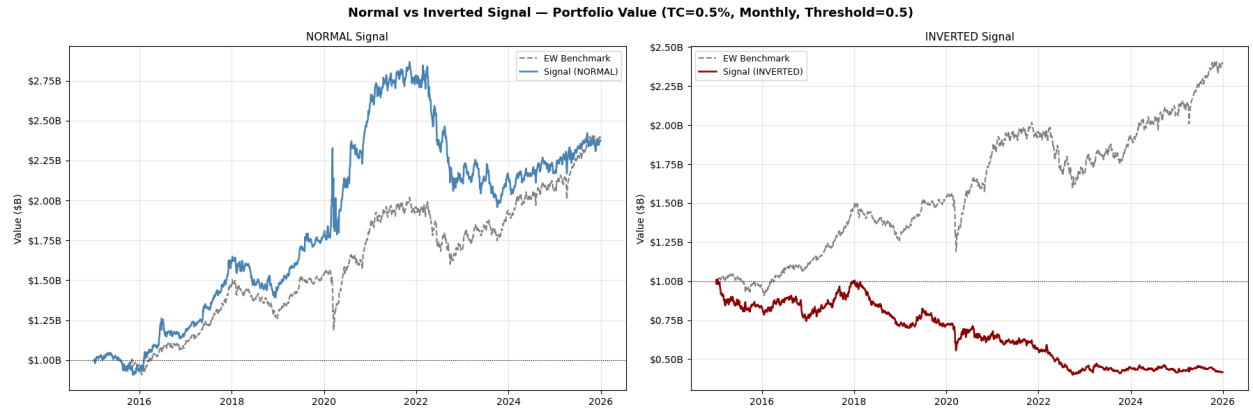


Figure 21: Comparison of Portfolio strategy vs. inverted decision portfolio

6.5 Binary Regime Comparison

Table 5: Portfolio Performance Metrics Comparison

Strategy	Total Return (%)	CAGR (%)	Volatility (%)	Sharpe	Max DD (%)	Calmar
Determinant	184.31	12.96	13.67	0.976	-13.96	0.928
Buy & Hold	166.21	12.09	22.06	0.620	-42.12	0.287
Equal Weight	162.58	11.92	17.29	0.729	-40.47	0.294
MV	43.93	4.34	20.79	0.313	-49.07	0.088
HRP	229.53	14.92	20.45	0.783	-40.91	0.365

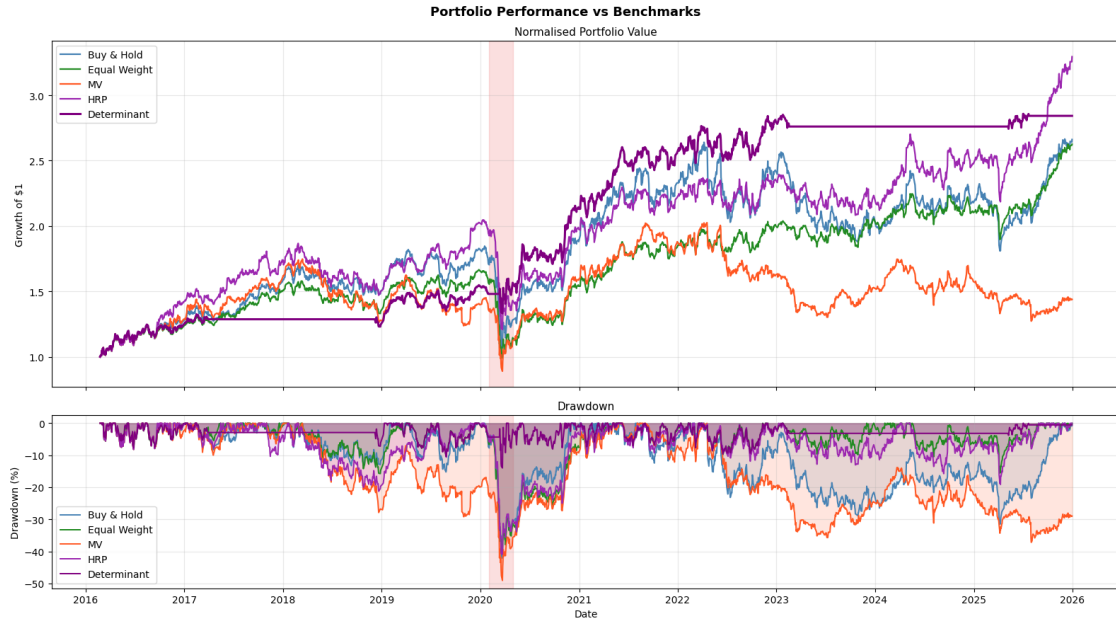


Figure 22: Portfolio performance comparison of all investment approaches

6.6 Weight Dynamics - Binary Regime



Figure 23: Weight dynamics comparison of all investment approaches

7 Conclusion and Outlook

We have shown that the Koherent-Interdependence-Korrelation-Operator (KIKO) is capable of extracting signals that can be used to improve portfolio decisions. Especially during volatile and bearish market environments, the interdependence structure (ISAAC) identified through the adaptive asset allocation scheme shows promising results. The determinant-based signal appears to capture meaningful changes in market structure and can be utilized to improve investment decisions compared to several standard benchmark approaches. In our GitHub repository, we additionally provide several extensions and variations of the proposed signal framework. However, we decided to keep the presented work at a digestible level, as the first additional attempts did not produce signals that were substantially more informative than those presented in this study. Another natural extension would be to test all possible rolling-correlation windows within a larger parameter study. For example, combinations of 2-day and 2-week windows, 1-day and 5-day windows, or various combinations of short-, medium-, and long-term signals could be investigated systematically. Due to time constraints, we were not able to conduct such an exhaustive parameter analysis. One drawback of the current approach appears to be its relative underperformance during calm or bullish market periods, where the binary determinant signal may lose momentum compared to steadily rising markets. Future investigations could therefore focus on improved portfolio construction techniques and more accurate partial hedging of investment positions. So far, we have not incorporated short-selling or leverage into our portfolio construction framework, as we aimed to benchmark a relatively simple and practical approach against standard portfolio implementations. Especially for large investment vehicles such as pension funds, one would need to carefully reconsider the feasibility of repeatedly buying and selling the entire portfolio. In such settings, transaction costs do not only play a minor role but can also create market impacts when large amounts of assets are bought or sold. From a risk-management perspective, the reduction in drawdowns appears to be one of the most promising features of the proposed methodology. Nevertheless, a more detailed investigation of risk measures like expected shortfall, tail-risk characteristics, and regime-dependent risk metrics could add significant value to the current framework. Beyond the statistical and econometric nature of the proposed approach, the current era of machine learning naturally suggests further extensions. More sophisticated machine-learning and deep-neural-network architectures could potentially incorporate the extracted determinant signals as additional features, thereby improving predictive performance and investment decisions.

We would also like to thank Professor Byers for his comments, support, and guidance throughout our Capstone Project. The fruitful discussions we had and the valuable insights he provided were instrumental in shaping this work. As a final remark, we would like to thank WorldQuant University for providing such a brilliant program. We learned a great deal, made new friends, engaged in valuable collaborations, and established the foundation for a new stage in our professional careers. Thank you very much - with sincere appreciation and respect,

KIKO - Kristian Boroz
ISAAC - Isaac Kiplimo Sum
I-WAN - Israel Twum Bosompem

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